An Introduction to Replication Research in Gifted Education: Shiny and New Is Not the Same as Useful

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Abstract

This methodological brief introduces readers to replication methods and their uses. Broadly defined, replication is the duplication of previously conducted research to verify or expand the original findings. Replication is particularly useful in the gifted education context because so much education theory and research are based on general education students and environments; the extent to which this prior work extends to gifted education contexts needs to be evaluated. The rationale, definition, implementation methods, and current state of replication in the social sciences generally, as well as in giftedness and creativity research specifically, are reviewed. Finally, guidance for conducting replications is provided, including recommendations on selecting important topics, engaging original authors, ensuring sufficient power, using appropriate measures and analyses, framing the manuscript, and submission planning.

Keywords

gifted education, replication, research methods

Increasing the theoretical and statistical rigor of research has garnered substantial attention in giftedness research (e.g., Plucker & Callahan, 2014b; Thompson & Subotnik, 2010). Using more sophisticated analyses allows researchers to assess program effects and student outcomes more accurately. However, neither is methodological rigor limited to advanced statistics, nor are statistics being universally correctly applied (Bakker & Wicherts, 2011). Moving beyond statistical rigor is important because relying on low *p* values will not assure that all "false" research findings are avoided (Ioannidis, 2005b; Pashler & Harris, 2012). Moreover, it is important for researchers and the general public to remember that even when publishing in peer-reviewed academic journals, researchers maintain incredible amounts of flexibility in how data are collected, analyzed, and interpreted. Such flexibility, or "researcher degrees of freedom," can lead to improbable and unrealistic results (see Simmons, Nelson, & Simonsohn, 2011, for an example of how listening to a particular song can "cause" people to be roughly 18 months younger than listening to another song). Such problems suggest that the findings from individual articles should be treated not as "true" but rather as individual pieces of evidence. One piece of evidence should not necessarily serve as the sole foundation for policy.

The diversity of definitions and approaches to measurement in giftedness and creativity research only adds to this flexibility found in psychological research. Given that how gifted students are identified and served can vary hugely across contexts, it is not always appropriate to assume that findings (e.g., whether a program is effective) from one study generalize to different programs. For example, when discussing the role of the researcher, Callahan and Moon (2007) warned,

> Unless research on [gifted education] is also conducted by independent researchers who do not have a vested interest in or potential bias about the interpretation of data collected, arguments that support the effectiveness of the models are weaker than arguments supported by outside, independent research studies. (p. 307)

One way to combat such freedom while also helping differentiate empirical facts from statistical anomalies is to replicate research findings. Replication can help identify when individual studies are outliers and when they follow general trends. Replication can also identify random errors and measurement errors, while also uncovering how researcher degrees of freedom affect the stated results.

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First articulated by Lykken (1968), replication is now typically broken into two primary categories: direct and conceptual (Schmidt, 2009). *Direct replications* seek to follow the original study's methods (e.g., similar participants, measures, and procedures) as closely as possible in order to test whether the original research results can be obtained again. *Conceptual replications* purposefully alter at least one component of the original study to test the underlying construct of interest. Without clear and detailed methods sections in research articles, replications can be difficult or impossible to conduct. Replications serve as a critical part of the scientific enterprise by helping control for issues caused by sampling error, artifacts, fraud, generalizability, testing the general underlying hypothesis of a previous study, or some combination of those issues (Schmidt, 2009).

Conceptual replications can help identify the degree to which a particular finding is broad or narrow. For example, many recent studies have examined whether previous findings in general education environments on the effect of being in environments with higher average ability extend to gifted participants in supplemental summer programs (e.g., Dai, Rinn, & Tan, 2013; Makel, Lee, Olszewski-Kubilius, & Putallaz, 2012; see related commentary by Plucker et al., 2004). A reanalysis of previously collected data (e.g., Plucker, 1999) is not a replication but is still a vital part of the scientific process. Reanalysis of data can reveal things as simple as calculation errors as well as more fundamental problems such as data manipulation and fraud, and reanalysis can provide new information by reanalyzing seminal data sets using modern statistical techniques.

Although not often highlighted as such, replication has a long history in giftedness research. Meta-analyses and research syntheses on the state of research of central constructs such as acceleration (e.g., Colangelo, Assouline, & Gross, 2004; Steenbergen-Hu & Moon, 2011) could not be written without numerous studies addressing acceleration questions. However, there is also a catch-22: Replication is good for the field but not necessarily for the individual researcher. This is because the current publication climate does not prioritize accuracy but instead emphasizes being the first to publish interesting and surprising results (Asendorpf et al., 2013; Makel & Plucker, 2014a, 2014b; Nosek, Spies, & Motyl, 2012).

Makel and Plucker (2014b) list eight systematic biases against the publication of replications. This list includes biases pertaining to conducting replications (e.g., bias against funding replications as well as against hiring/promoting individuals who conduct replications) as well as publishing replications (e.g., a bias against even submitting replications, as well as biases built into journal editorial policies). Makel and Plucker also note that a bias equating novelty with creativity exists in the current academic research environment. Ironically, this prioritization of novelty over accuracy fosters even more need for replication to help identify which findings not only are the most robust but also actually exist. Indeed, given the researcher degrees of freedom discussed above and other methodological problems including low power (Schimmack, 2012) and insufficiently narrow confidence intervals (e.g., Kelley & Maxwell, 2003), some have suggested that "most published research findings are false" (Ioannidis, 2005b, p. e124).

Despite their importance in building a reliable, valid research base, replications are often viewed as having low prestige because they do not necessarily contribute new, blockbuster findings. However, they are not without value. If research findings are to be used to help set educational policy, verification of results becomes all the more important. For example, many believe that education policy should not be based on the results of one study. This is relevant because individual studies can be based on data from only one school or even one classroom. Some schools do not have a gifted program, and if they do, there are vastly different practices for identification and service (e.g., McClain & Pfeiffer, 2012), which necessitates separate evaluation. For example, one recent evaluation of a gifted program (Card & Giuliano, 2014) reported vastly different effects of participation based on how students were selected for participation (i.e., based on IQ scores or high achievement test scores).

If the topic is of sufficient importance and the replication is of high quality, it need not doom the replication research to obscurity. Indeed, replications have been published in the most prestigious academic journals (e.g., Kristensen & Bjerkedal, 2007). Moreover, replications get cited at relatively high rates (Makel & Plucker, 2014b; Makel, Plucker, & Hegarty, 2012). Additionally, given how many research findings and theories are developed using general student populations, giftedness is particularly apropos for replication research; it may be that giftedness differentiates students in such a way that research conducted with general student populations is not applicable. The best way to assess generalizability of findings from the general population to gifted students is through replication.

In a review of highly cited medical research, Ioannidis (2005a) found that only 44% of replications reported results similar to the original study. Studies that relied on small sample sizes and that were not randomized (two common features of giftedness and creativity research) were most likely to lead to contradictory results when replicated. Other researchers have found even lower replication success rates in medical research (e.g., Begley & Ellis, 2012; Prinz, Schlange, & Asadullah, 2011). If fields commonly associated with large samples and methodological rigor do not replicate results, what is to be said about fields known for small samples (e.g., giftedness research)? Two recent reviews of the top 100 psychology and education journals have reported that although very few articles are published as replications in either field (1.07% and 0.13%, respectively), the majority of those studies successfully replicate the original findings (Makel, Plucker, et al., 2012; Makel & Plucker, 2014b). However, in both fields, over half the replications are

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Table 1. Replication Rates in Giftedness and Creativity Journals.

Journal name	Publication years	No. of replications	Total articles	Replication rate, %
Gifted Child Quarterly	1957-2012	4	1,945	0.21
Journal for the Education of the Gifted	1992-2004	3	325	0.92
High Ability Studies	2000-2011	0	218	0.00
Psychology of Aesthetics, Creativity, and the Arts	2009-2012	5	145	3.45
Creativity Research Journal	1994-2012	10	685	1.46
Journal of Creative Behavior	1967-2012	3	1,278	0.23

conducted by the same research team as the original study. As Callahan and Moon (2007) warned in their discussion of gifted education research, this is relevant because replications with authorship overlap are more likely to be successful than replications conducted by nonoverlapping researchers (Makel, Plucker, et al., 2012; Makel & Plucker, 2014b).

An analysis of six giftedness and creativity journals conducted for this methods brief (Table 1), using the same methods as in the previous studies, revealed that 0.5% of published articles are framed as replications. So few replications being published engenders a policy environment where decisions have to be made with little or no information on what actually creates best practices (see Plucker & Callahan, 2014a, 2014b). With small samples often relying on different operationalizations of the primary constructs (i.e., giftedness or creativity) from relatively disparate contexts, it can be difficult to conclude with confidence how far a particular research finding will generalize.

Replication is not limited to strict experimental research either; qualitative research can and should also be replicated (Golden, 1995). Additionally, action research (e.g., Tomlinson, 1995) is also a good avenue for replication research; it naturally tests the fit of previously established models in a particular educational environment.

Ways to Partition Replications

Replications are usually deemed as either a success (i.e., getting similar results as the original research) or a failure (i.e., getting differing results from the original research). A successful direct replication helps verify the original finding, whereas a successful conceptual replication helps build on the original findings. However, this is usually determined merely by direction and not by anything as sophisticated as a point estimate or a measure of magnitude (cf. Simonsohn, 2014; Verhagen & Wagenmakers, 2014). Failed replications, on the other hand, can be more difficult to interpret. Failing to *directly* replicate a finding may imply that the original finding is flawed, that the replication is flawed, both, or neither. Similarly, failing to *conceptually* replicate may imply that the original finding is flawed, that it does not generalize to the new sample/method in the replication, that the replication is flawed, some combination thereof, or none of the above.

For example, if the measure used is changed as well as the procedure for collecting data, a failure to replicate does not imply that the original findings are false, nor does it necessarily inform the reader about the merits of the new measure or the new procedure. A problem with either could have led to the failure to replicate the original findings. Some (e.g., Pashler & Harris, 2012) have suggested that direct replication of controversial findings should be the initial focus, with conceptual replication following (to avoid conflating methodological factors with contextual factors). Regardless, if failed replications are not published, then boundary conditions of when and where results are applicable will not be revealed (Giner-Sorolla, 2012) and others are doomed to keep repeating mistakes. Such boundary conditions may be particularly relevant to giftedness contexts given that most education research and theories are conducted and developed with a general student population.

Overview of How to Conduct a Replication

Two examples of replications from the giftedness field were conducted by Lee, Cramond, and Lee (2004) and Simonton (2008). Both replicated previously published work with the goal of testing whether previous findings extend to different populations.

Conceptually replicating Tannenbaum's (1962) classic study on U.S. anti-intellectualism, Cramond and Martin (1987)[1] asked educators instead of adolescents (an attempt to generalize to a different population) their views of hypothetical students based on varying levels of student academic ability, effort, and athleticism. Lee et al. (2004) further replicated both studies with a Korean educator sample (thus further expanding the population tested). In both cases, the replicating studies helped generalize the original results to different populations (to educators and educators from a different culture, respectively). In this particular case, such generalizations help inform researchers and educators that anti-intellectualism (particularly when the students are studious and nonathletic) is not limited to adolescents but pervades the education system, and it is not limited to the United States but also exists in a culture that highly values educational success. Knowing this is important, if any intervention

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to combat anti-intellectualism were to be planned, teachers cannot necessarily be relied on to help combat student anti-intellectualism.

Simonton (2008), on the other hand, conceptually replicated Cox's (1926) classic historiometric study that retrospectively investigated the extent to which eminent adults could have been identified as gifted while they were adolescents. Using a sample of eminent African Americans, Simonton successfully replicated the original underlying finding (eminent adults showed signs of high achievement in adolescence) with a different population and by using slightly different methods to identify eminent participants.

Both articles changed multiple aspects of the study: in Simonton's (2008) case, the population being analyzed and the selection methods of genius. In Lee et al.'s (2004) case, the population being studied (Korean vs. American), the role participants played in education (preservice and in-service teachers instead of students), as well as changing ability to be self-reported rather than assessed were changed. Although making multiple methodological changes is common, it can also muddy interpretation of results, particularly when results differ from the original. The preferred systematic solution is the manipulation of single variables at a time. However, the authors are hardly to blame for making multiple changes given the current publication climate that seems to require both novelty and statistical significance in publication (e.g., Fanelli, 2010; Makel, 2014).

Selecting a Replication Topic

Two important questions need to be considered when selecting a replication topic. First, what needs replicating? Importance can be measured in many ways, but selecting the right research findings to replicate can be helpful. Some facets to consider include the following:

- Has the original research been highly cited? Citations suggest a topic of high interest to other researchers. Replications for highly cited work may be more relevant than replications of work that has not been highly cited.
- Has the original research been implemented in the classroom? If yes, practitioners have expressed interest in the finding. Having a strong research base assessing if, when, and how the research is applicable would help inform educators which practices are effective in particular contexts.
- Was the original research conducted only in general education contexts and not in gifted education contexts? If so, it may be inappropriate to assume that the same findings hold true for gifted students or in all gifted education contexts.
- Have contextual factors sufficiently changed to call the original findings into question? Such changes could result from changes in gifted identification practices, policy, demographic changes, or technological advancement.
- And, perhaps most pessimistically, is there concern that the original findings are fraudulent? Just as an individual study should not be considered a monolith of truth, a single failed replication should not be considered a fatal blow to a previously published finding. However, replications can help alert the field about the strength (or fragility) of previously published findings (for a more in-depth discussion of replications, see Makel & Plucker, 2014b; Schmidt, 2009; as well as special sections in *Perspectives on Psychological Science*: "Special Section on Moving Toward a Cumulative Science," 2014; "Special Section on Replicability in Psychological Science," 2012).

Second, what are the goals of the replicators? Are they seeking to check the data or the underlying construct? As discussed previously, if a finding has not been directly replicated, attempts to conceptually replicate the research may be premature. However, if a conceptual replication is planned and if two factors seem questionable, changing both simultaneously can lead to unclear interpretation; replicators may want to have multiple conditions where each factor is manipulated individually as well as one where both are changed. This will be particularly useful if one (or both) conditions fail to replicate the original findings.

Some replications can contain aspects of both direct and conceptual replications. For example, in an article studying changing gender gaps in the extreme right tail of mathematical and verbal ability, Wai, Cacchio, Putallaz, and Makel (2010) directly replicated the highly cited finding of a 13:1 ratio favoring males in the top .01% of mathematics in the early 1980s (Benbow & Stanley, 1980, 1983). However, the newer study also conceptually replicated the original finding by showing that the ratio rapidly declined in subsequent years and by using additional tests (i.e., the ACT) and academic domains (science, verbal ability, and writing). This replication was also independently successfully replicated (Olszewski-Kubilius, & Lee, 2011). Successful direct replication can help establish credibility for a subsequent conceptual replication that is expected to have findings that differ from the original research.

Conducting a Replication

Replication research can be a wonderful opportunity for student assignments in research methods classes (Frank & Saxe, 2012; Grahe et al., 2012). Replication assignments can help students gain experience conducting research without the burden of developing innovative questions or methods. It allows them to practice focusing on how to conduct sound research. One reviewer correctly noted that high school students, with the proper mentoring, could conduct replications in an effort to learn social science methodology and the content of the study.

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When planning a replication, replicators should not be afraid to engage original authors when planning their study (although all is not lost if that is not possible). Even well-intentioned authors may not have been able to communicate every subtle detail of their research methods in their final publication. Contacting the original authors, stating the intent to replicate, and asking for additional information/detail may help avoid potential problems along the way. It can also help frame the replication as collegial instead of threatening (Kahneman, 2014). Some authors may react poorly (especially if the replication is framed as an attempt to discredit the initial research findings), but replication attempts need not be viewed with hostility. Indeed, it can be quite flattering to have a work considered so important that someone else wants to spend their time investigating the same ideas (Spellman, 2012)! If the original authors do not reply (a month to reply as well as a reminder e-mail should be sufficient), replicators can still include mention of the attempt in their methods section. This communicates to readers that an attempt was made to obtain all relevant information on the original study.

If conducting a quantitative replication, having sufficient power/sample size is of utmost importance (Bakker, van Dijk, & Wicherts, 2012; Cohen, 1962, 1990; Schimmack, 2012). Having sufficient statistical power is necessary for precise research results (Cumming, 2014; McBee & Matthews, 2014). For example, a study with an insufficient sample size could yield an ephemerally large effect size of 1.0. But if the 95% confidence interval of that effect size ranges from 0 to 2.0, the result becomes uninterpretable and effectively useless (for additional discussion, see Cumming, 2014; McBee & Matthews, 2014). In comparison, a comparable study with a larger sample could yield an effect size of "only" 0.5 but with a 95% confidence interval of 0.4 to 0.6. This study would provide a far stronger case for an effect, even though the estimated effect size appears to be half that of the smaller sample. There are many free power estimate calculators available online, including G*power 3 (Faul, Erdfelder, Lang, & Buchner, 2007), to help determine the needed sample size based on the estimated size of the effect (which can potentially be obtained from the research being replicated).

Power problems (and resulting imprecise results) help contribute (along with reviewer and editor mandates) to over 80% of all scientific articles having significant results, with psychology having over 90% of its articles having statistical significance (Fanelli, 2010; Sterling, Rosenbaum, & Weinkam, 1995). Such findings indicate publication bias, which is troublesome when considered in the context of failed replications being de facto barred from being published. If only positive findings are reported, how do we know if 20 other research teams attempted to conduct similar research and failed to obtain similar results?

Measures. As discussed above, direct replications should use the same methods and measures as the original study. If the measure has not been published or publically shared, this is where contacting the original authors may be necessary. If a different measure will be used, an explanation of the differences between the new and original measure is necessary to justify why the new measure is worth using. In some cases, changing the participant group may necessitate a change in the measure. For example, there are different measures of self-concept based on the age of the participants. Similarly, a qualifying measure used to identify gifted students in elementary school may be inappropriate when identifying gifted high school or college students.

Participants. How are participants different from the original sample? Simply having a sample of students who have been identified as gifted may not necessarily provide an equivalent sample to the original study. The practices used to identify each group as gifted must be included as well as whether they are academically and demographically similar to the original sample. Having a sample that differs from the original sample is fine, but it would likely constitute a conceptual replication.

Analysis. If possible, the analyses from the original study should be used for both direct and conceptual replications. If the method of analysis is part of what is being questioned, it may be of value to conduct the analysis using both the original method as well as the new method to illustrate clearly the relevance of the difference.

Determining Success or Failure

As discussed above, whether a replication is a considered a success or failure has traditionally been determined by simply looking at whether *p* values trend in the same direction as the original results. Such methods do not take into account whether the effects sizes are similar. Several newly developed methods have been proposed to help quantify the success or failure of replication attempts (e.g., Simonsohn, 2014; Verhagen & Wagenmakers, 2014), but the calculations involved go beyond the scope of this methods brief. Broadly, Simonsohn (2014) proposed that replicators reference the original sample size and determine what would constitute a reasonable small but detectable effect. With this as a benchmark, replicators can then conduct their replication to see if their result is smaller than this small effect. If so, the replication is considered to have failed to replicate the original findings (for similar estimates of sample size needed for sufficiently narrow confidence intervals on effect sizes, see Kelley & Preacher, 2012).

Verhagen and Wagenmakers (2014) describe several additional ways replications can be assessed (while also pointing out their flaws). These methods include conducting a small-scale meta-analysis based on the original and replicated findings and several methods relying on variations of Bayesian hypothesis tests that assess whether there is a difference

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between the effect sizes of the original and replicating research (Bayarri & Mayoral, 2002), whether the combined effect of all studies provides evidence for an effect (Rouder & Morey, 2011), and whether the effect in the replication is similar to the original effect reported (Verhagen & Wagenmakers, 2014; sample R code for calculations is included in text).

Replication Manuscript Preparation

The authors of the original research are likely reviewers, so replicators should avoid ad hominem attacks at all costs and be sure to keep focused on the research process and research findings. Replications are about ideas, not individuals. Moreover, being overly critical of the original research may feel like a good way to highlight the importance of the replication, but it may also cause the original authors to become defensive (for an example of how replication can devolve, see Meyer & Chabris, 2014).

Several journals (e.g., *Journal of Advanced Academics, Perspectives on Psychological Science, Psychological Science*) have had editorials announcing the acceptance of replications (Eich, 2014; McBee & Matthews, 2014). Replicators should be sure to refer to the editorial policies of journals before beginning to write (or, in the case of *Perspectives on Psychological Science*, before the replication is conducted!). Some direct replications can be quite short (e.g., LeBel & Campbell, 2013) whereas some conceptual replications require substantial explanation for what has been altered (e.g., Makel, Lee, et al., 2012). Regardless, very few individuals believe that any one study can provide the definitive answer to a particular issue, so replicators should be sure to avoid framing the replication as anything other than providing additional information to the issue. This is not saying that the new results should be trivialized, but the replication findings will not likely end the conversation. Providing evidence that the conversation needs to continue is a sufficiently important intellectual contribution.

On the other side of the coin, some editors may still be predisposed against publishing replications. Thus, checking with editors prior to submission may save replicators some time and effort. Trying to convince an editor of the importance of replication is commendable but may substantially lengthen the time until the replication manuscript is published.

Concluding Thoughts

The pursuit of science is not to pursue novelty but rather to differentiate fact from fiction. Replications offer gifted education researchers a clear path to understand better the extent to which previously published findings/theories from other educational contexts, using other student populations (or even other samples of students identified as gifted using a different set of criteria), extend to gifted education environments and students. This path allows researchers and evaluators more precise information on practices that are (or are not) effective with gifted students and away from practices based on "armchair best thinking" (Callahan & Moon, 2007, p. 305). This is particularly important because research that does not focus specifically on gifted students may overlook potential confounds or exceptions relevant to gifted students.

A group of social psychologists from 36 independent labs across the globe recently collaborated to conduct a large-scale replication of 13 classic and "hot topic" findings (Klein et al., 2014). This "many labs" collaboration preregistered their hypotheses, shared data from a 15-minute questionnaire sent to over 6,000 participants in 12 countries, and successfully replicated 10 of the 13 studies. Not all replications have to be so large in scope, but this could serve as a model for how researchers studying gifted education and high performance can work together for the benefit of the field. By conducting such collaborations, researchers can overcome concerns of self-serving biases (i.e., only replicating their own work) while strengthening the credibility of the field as a whole.

Replications are merely one of numerous solutions (e.g., reanalysis of data, preregistering hypotheses, and publishing *studies* of interest rather than *results* of interest) for increasing the rigor of research. But replication is critical for the development of the field because although fraud and error may eventually be revealed in an absence of replication, in the intervening time, research projects and educational policies will be based on faulty results. Replication can help identify errors more quickly while also potentially helping verify and clarify previous results, leading to a more effective and more highly respected body of gifted education research.

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Note

1. Although this is also a replication, it did not come up in a search for replications because no form of the word *replication* is used in the article.

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